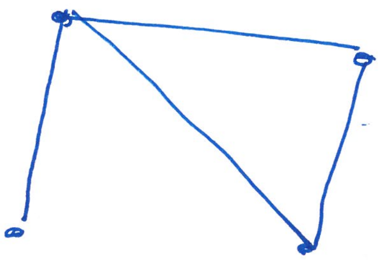
LECTURE 36: MATH1061/7861
TOPICS: WALKS, TRAILS AND CIRCUITS
DATE: THURSDAY MAY 21, 2026

Learning objectives:

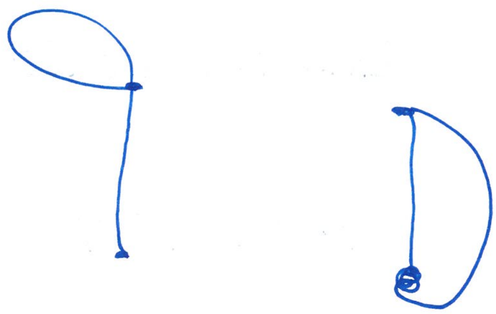
(1) Reinforce basic definitions and notation in graph theory.

(2) Understand when a graph contains an Eulerian circuit or an Eulerian trail.

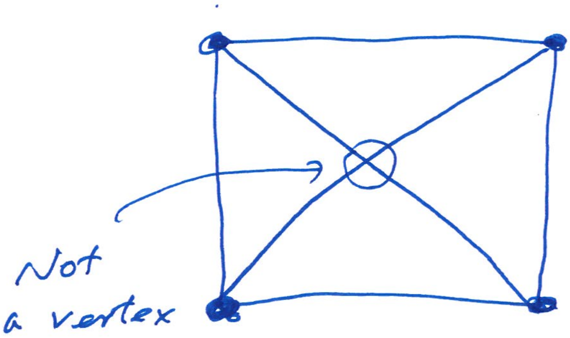
Connected vs. disconnected Graph.



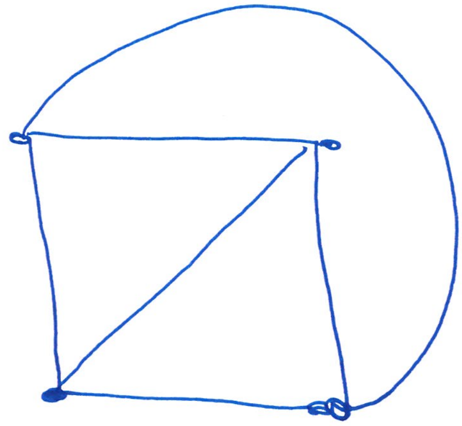
Connected.



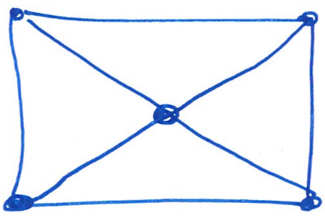
Disconnected



Are equivalent



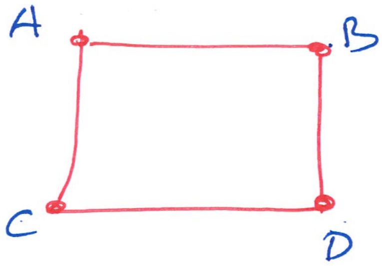
Different from



Eulerian Circuit :

Eulerian Trail

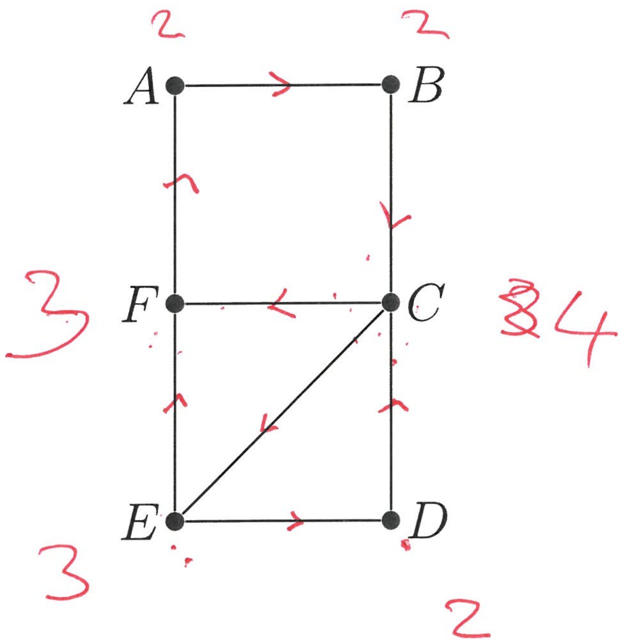
visit every edge exactly once.



ABDCA
Eulerian circuit.

Q1: Determine whether the following graphs have a Eulerian circuit or Eulerian trail.

(a)

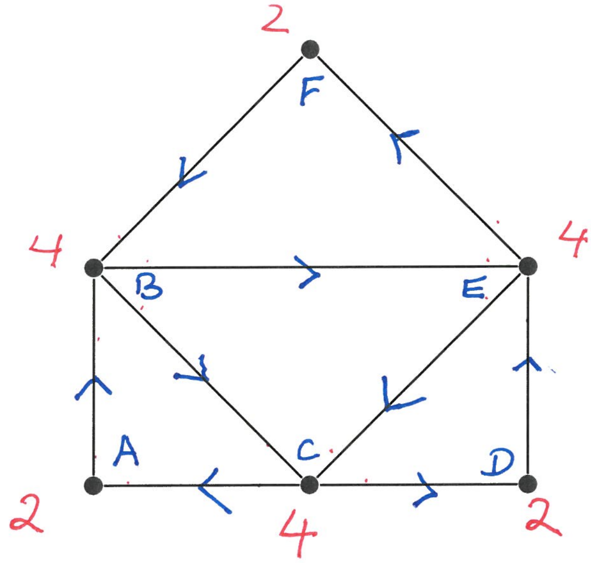


E F A B C ~~B~~ E D C F ← Eulerian Trail.

No Eulerian Circuit

Theorem: A connected graph has Eulerian Circuit if and only deg of every vertex is even.

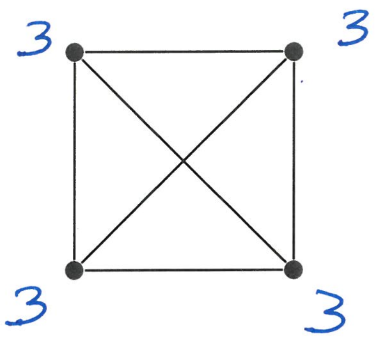
(b)



Eulerian Circuit: F B C A B E C D E F

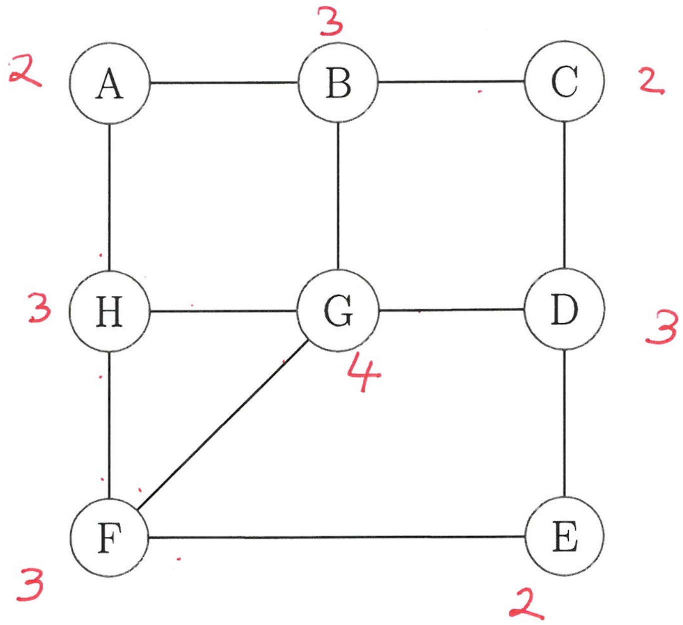
C A B E C D E F B C

(c)



No Eulerian circuit or Eulerian Trails.

Question: A network engineer needs to test the integrity of every physical link in a local cluster. The testing packet must traverse each link exactly once to minimize overhead. Is it possible to do this with the following network design? NO.



There are 4 vertices of degree 3.
This implies no Eulerian trail.

Hamiltonian circuit (or trail)
↳ Passes each vertex exactly once.

$\boxed{\text{Traveling Salesman Problem}}$